Development of a Solution for Situation-Awareness based on the Analysis of Speech and Environmental Sounds [Call Analysis]

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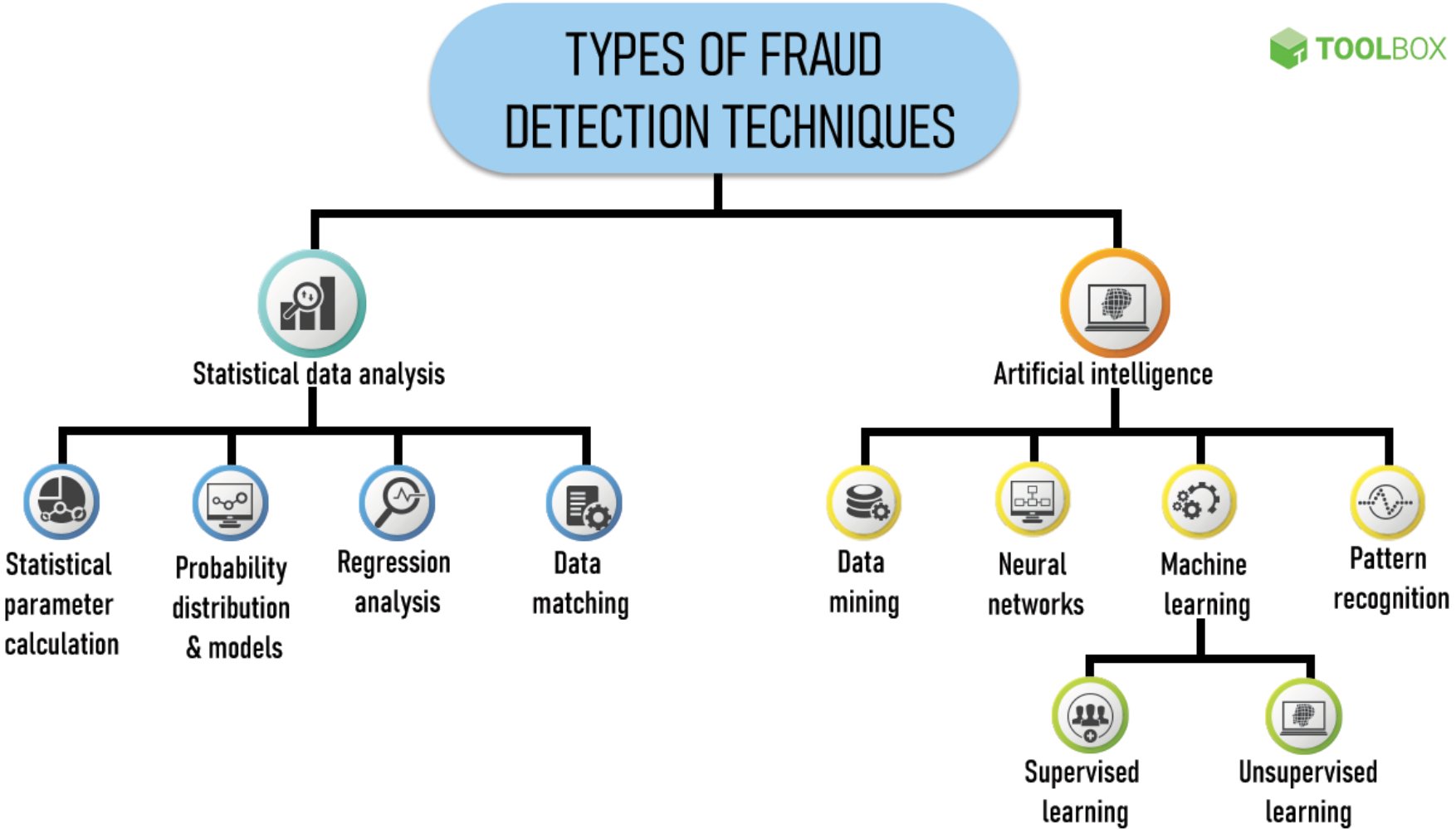
Sungkyunkwan University



Contents List

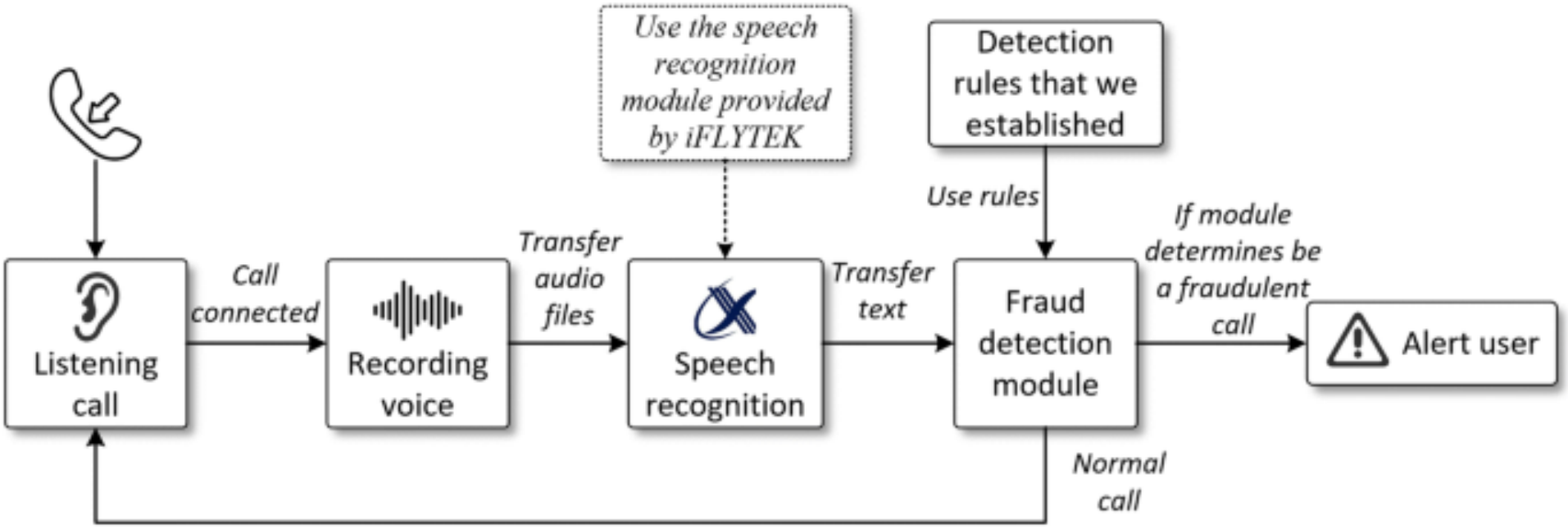
1. Fraud Detection
2. LISTEN's Background
3. LISTEN's Key Modules
   A. Gender Recognition
   B. Age Prediction
   C. Speech Emotion Recognition
   D. Fusion and Reasoning
4. **SER Patent**

1. Fraud Detection



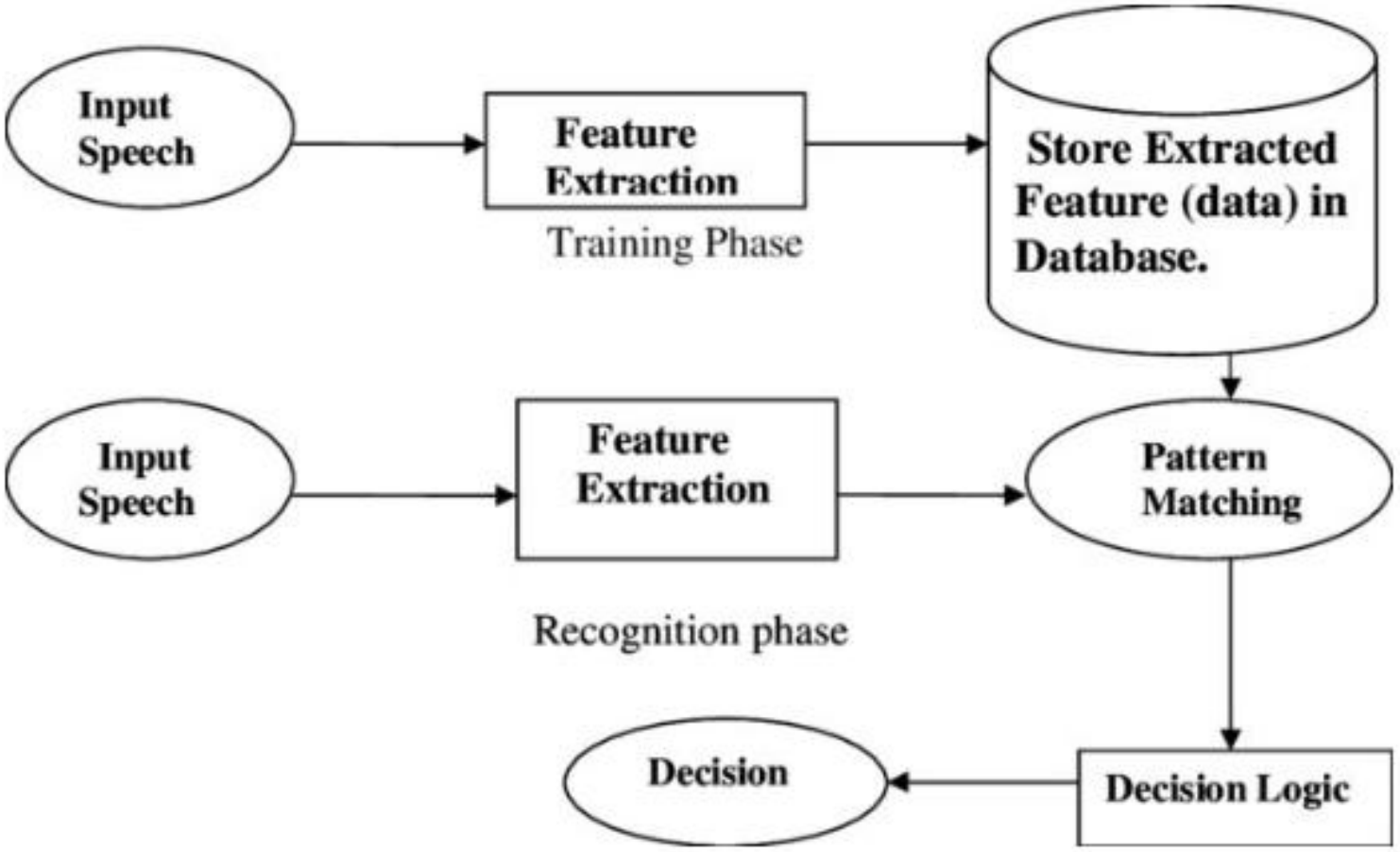
Zhao, Q., Chen, K., Li, T., Yang, Y., & Wang, X. (2018). Detecting telecommunication fraud by understanding the contents of a call. *Cybersecurity*, *1*(1), 1-12.

1. LISTEN Project's Motivation/Background



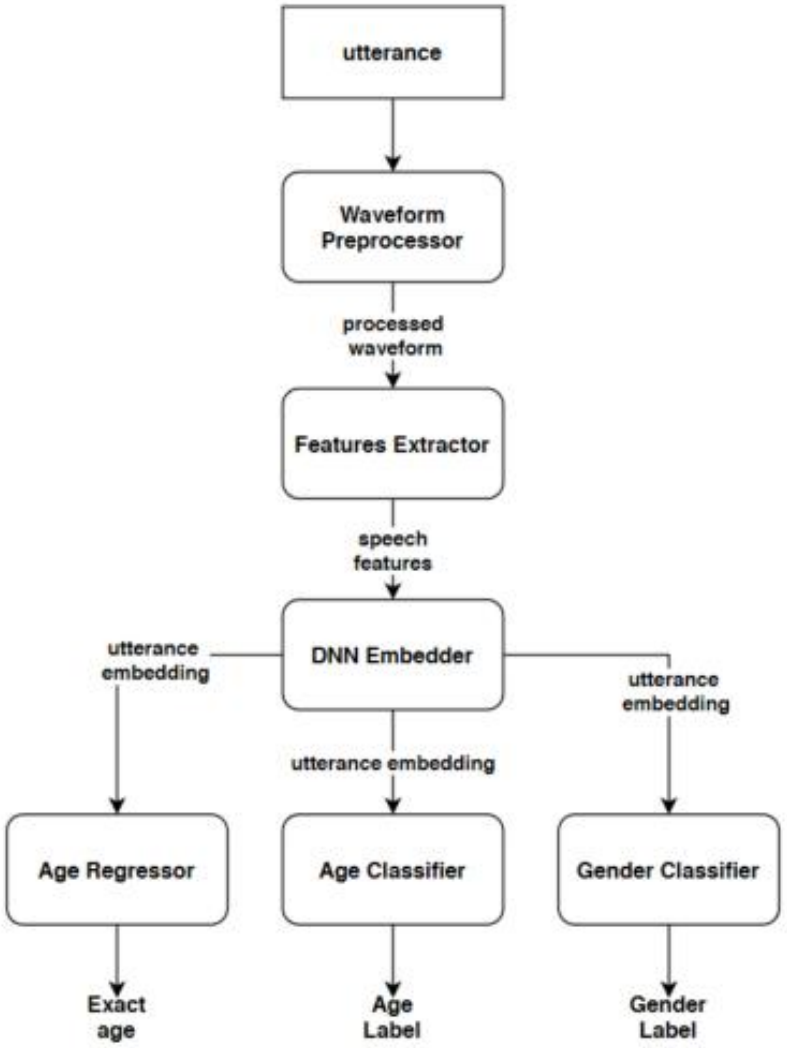
Zhao, Q., Chen, K., Li, T., Yang, Y., & Wang, X. (2018). Detecting telecommunication fraud by understanding the contents of a call. *Cybersecurity*, *1*(1), 1-12.

3.A Gender Recognition



Ahmad, J., Fiaz, M., Kwon, S. I., Sodanil, M., Vo, B., & Baik, S. W. (2016). Gender identification using mfcc for telephone applications-a comparative study. International Journal of Computer Science and Electronics Engineering (IJCSEE) Volume 3, Issue 5 (2015) ISSN 2320–4028 (Online)

3.B Age Prediction



Kwasny, Damian, and Daria Hemmerling. "Gender and age estimation methods based on speech using deep neural networks." *Sensors* 21.14 (2021): 4785.

3.C Speech Emotion Recognition

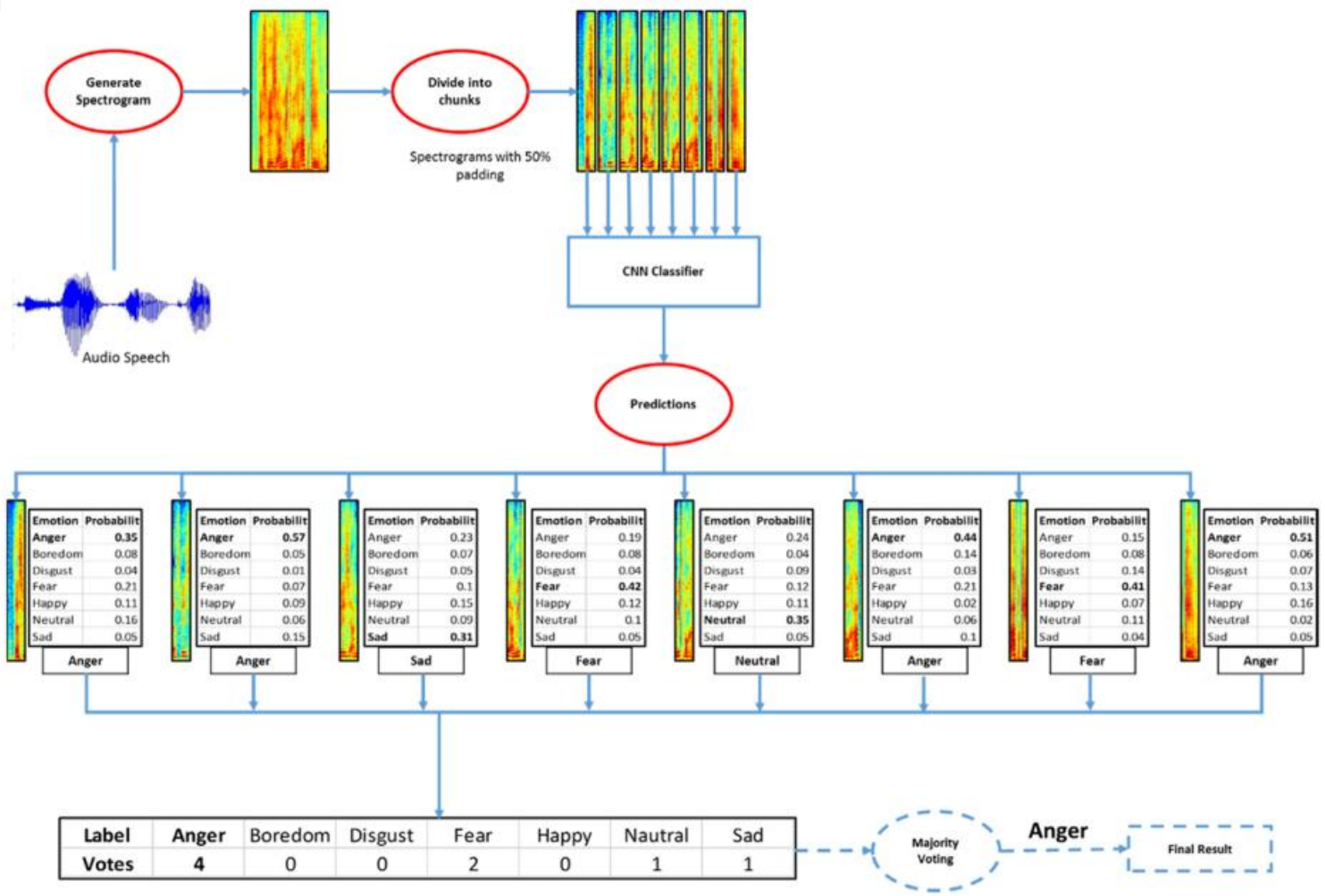


Fig. 4 Schematics of the proposed framework

Badshah, Abdul Malik, Nasir Rahim, Noor Ullah, Jamil Ahmad, **Khan Muhammad**, Mi Young Lee, Soonil Kwon, and Sung Wook Baik. "Deep features-based speech emotion recognition for smart affective services." Multimedia Tools and Applications 78, no. 5 (2019): 5571-5589.

3.D Fusion and Reasoning

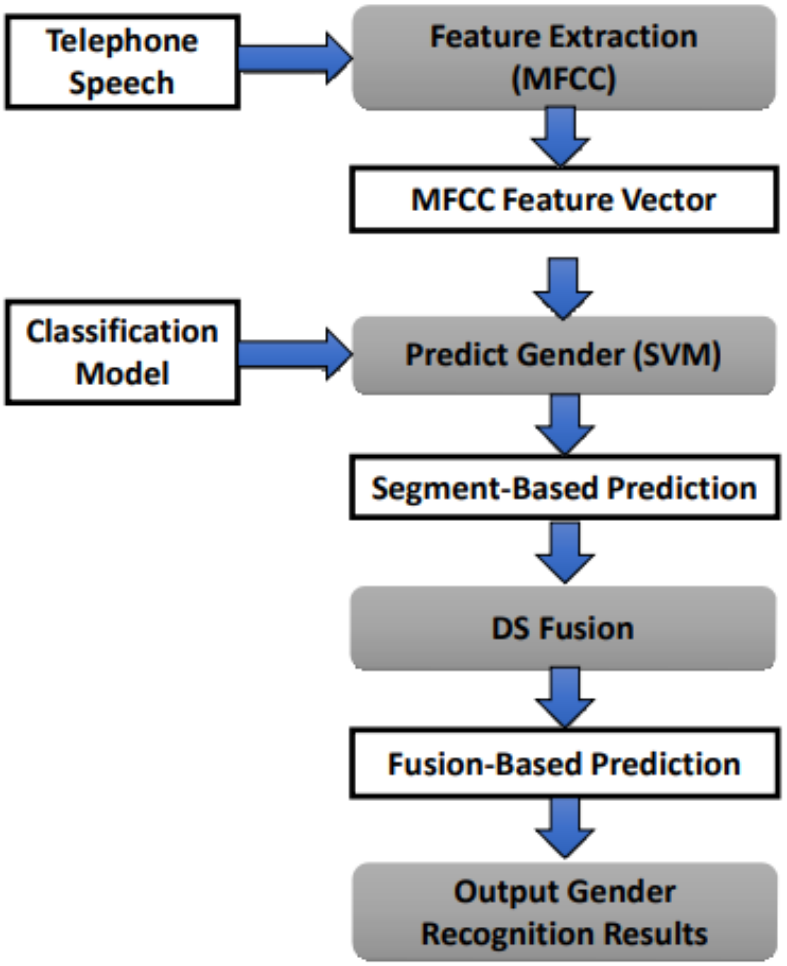


Fig. 1. Proposed reasoning framework for gender recognition

Ahmad, J., **Muhammad, K**., Kwon, S. I., Baik, S. W., & Rho, S. (2016, February). Dempster-Shafer fusion-based gender recognition for speech analysis applications. In *2016 International Conference on Platform Technology and Service (PlatCon)* (pp. 1-4). IEEE.

LISTEN Contributions (Patents)

Patent 1
SITUATION JUDGMENT SYSTEM AND METHOD BASED ON VOICE/SOUND ANALYSIS, KIPRIS 2016.
Applicants:
1. KOREA INSTITUTE OF SCIENCE AND TECHNOLOGY
2. TRIPHOS
3. WISENUT, INC.
4. INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY
5. Powervoice Co., Ltd.
6. IUCF-HYU (Industry-University Cooperation Foundation Hanyang University)

Patent 2
APPARATUS AND METHOD FOR GENDER RECOGNITION OF SPEECH DATA, KIPRIS, 2016
Applicant: INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY

Patent 3
APPARATUS AND METHOD FOR GENDER RECOGNITION OF SPEAKER, KIPRIS, 2016.
Applicants: INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY

Patent 4
APPARATUS AND METHOD FOR AGE GROUP RECOGNITION OF SPEAKER, KIPRIS, 2017.
Applicant: INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY

Patent 5
APPARATUS AND METHOD FOR RELIABILITY MEASUREMENT OF SPEAKER, KIPRIS, 2018.
Applicant: INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY

Patent 6
APPARATUS AND METHOD FOR SPEECH EMOTION RECONGNITION USING A REASONING PROCESS, KIPRIS, 2019.
Applicant: INDUSTRY ACADEMY COOPERATION FOUNDATION OF SEJONG UNIVERSITY

LISTEN Contributions (Research papers)

Ahmad, J., Fiaz, M., Kwon, S. I., Sodanil, M., Vo, B., & Baik, S. W. (2016). Gender identification using mfcc for telephone applications-a comparative study. International Journal of Computer Science and Electronics Engineering (IJCSEE) Volume 3, Issue 5 (2015) ISSN 2320–4028 (Online)

Ahmad, J., **Muhammad, K**., Kwon, S. I., Baik, S. W., & Rho, S. (2016, February). Dempster-Shafer fusion-based gender recognition for speech analysis applications. In *2016 International Conference on Platform Technology and Service (PlatCon)* (pp. 1-4). IEEE.

Badshah, Abdul Malik, et al. "Divide-and-conquer based ensemble to spot emotions in speech using MFCC and random forest." The 2nd International Integrated Conference & Concert on Convergence (2016).

Badshah, A. M., Ahmad, J., Rahim, N., & Baik, S. W. (2017, February). Speech emotion recognition from spectrograms with deep convolutional neural network. In *2017 international conference on platform technology and service (PlatCon)* (pp. 1-5). IEEE.

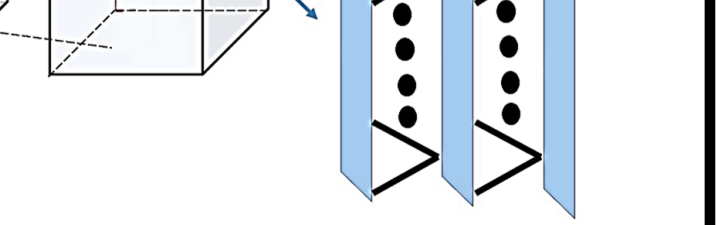
Ahmad, J., Sajjad, M., Rho, S., Kwon, S. I., Lee, M. Y., & Baik, S. W. (2018). Determining speaker attributes from stress-affected speech in emergency situations with hybrid SVM-DNN architecture. *Multimedia Tools and Applications*, *77*(4), 4883-4907.

Badshah, Abdul Malik, Nasir Rahim, Noor Ullah, Jamil Ahmad, **Khan Muhammad**, Mi Young Lee, Soonil Kwon, and Sung Wook Baik. "Deep features-based speech emotion recognition for smart affective services." Multimedia Tools and Applications 78, no. 5 (2019): 5571-5589.

4. SER Patent
Apparatus and Method for Speech Emotion Recognition using a Reasoning Process

Main Contents

1. Overall framework of our invention
2. Structure of the proposed feature extraction method
3. Co-existence of emotions for speech segments
4. Computing scores for previous N samples of a speech signal
5. Resolving conflict between conflicting emotions
6. Reliability-Weighted fusion for confidence values of three different sources
7. Maintaining emotional history for a speaker



indicate long-term features.

Prediction Confidence is measured using the max probability value.

nfidence
el?

Medium

High

g Emotions
and happy)
?

Yes

Resolve Conflict

Compute emotion co-existence score

Reliability-Weighted Fusion

Compute scores for previous N samples

Kr

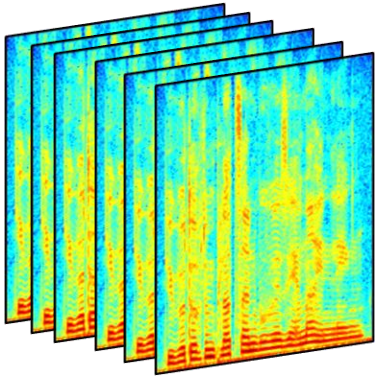
Emotion 1	Emo
Sad	Neu
Happy	Neu
Sad	Ang
Happy	Ang
Fear	Hap
.....	.....

Emotion

Speech segment	Emo
S1	Happy
S2	Happy
S3	Neutra

2. Architecture of the Proposed Classification Method

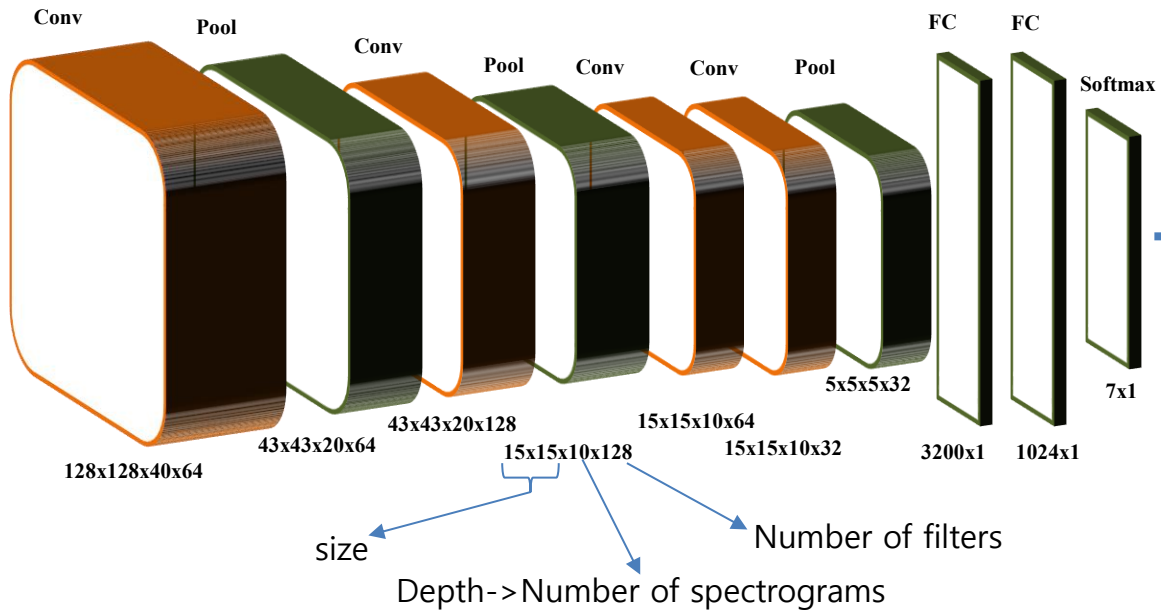
Input



Sequence of spectrograms of size M×N×D

M: Number of rows
N: Number of columns
D: Dimension

3D CNN Model



Stack of convolution and pooling layers, followed by Softmax classifier

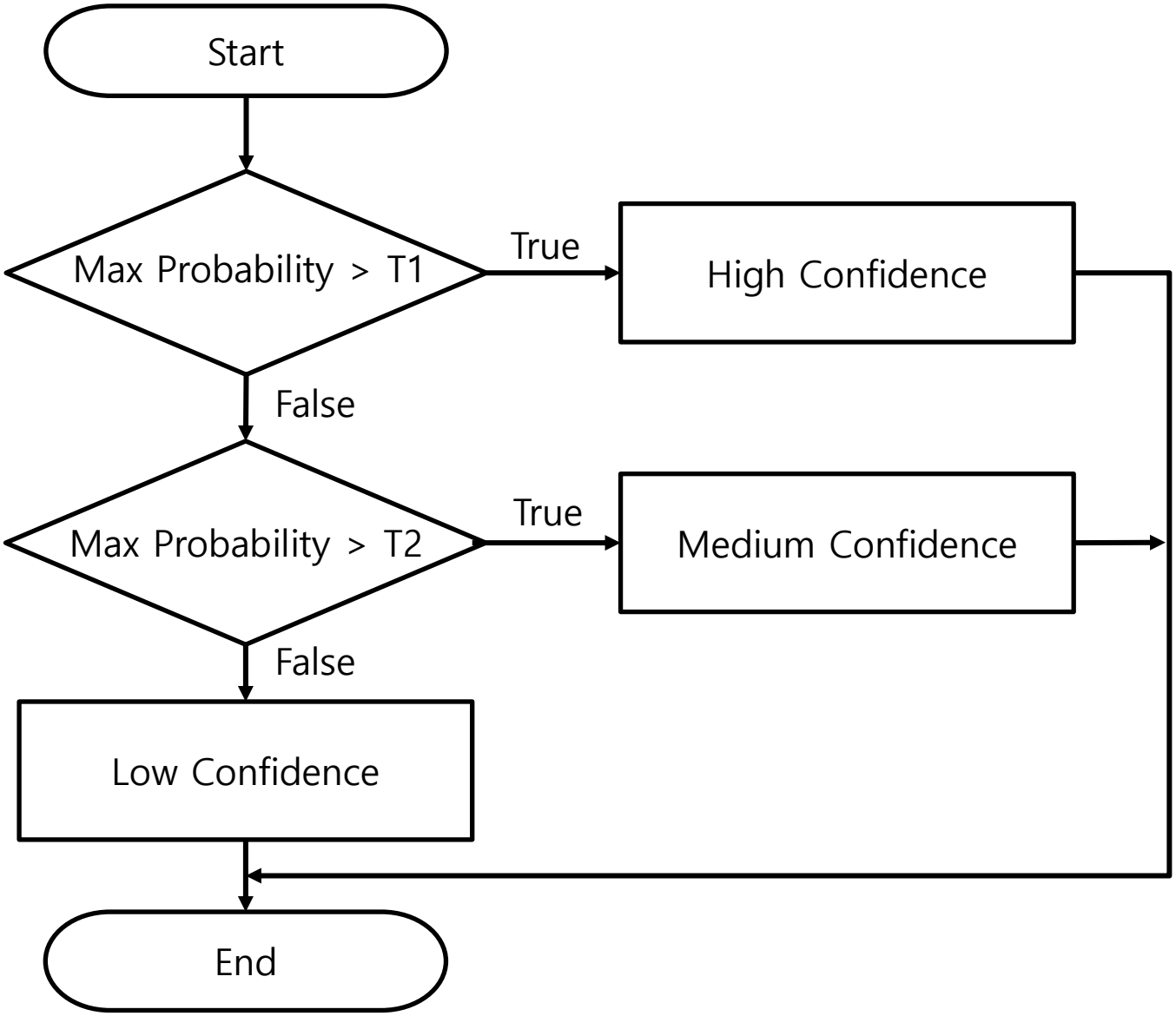
Output

Emotion	Probabilities
😐	0.12
😊	0.2
😕	0.05
😟	0.12
☹	0.4
😠	0.05
😖	0.06

Recognized emotions

Three-dimensional convolutional neural networks (3D-CNN) were first used for human action recognition [1].

Checking Confidence Level



3. Illustrating the co-existence of two emotions for speech segments

Emotion 1	Emotion 2	Probability
Happy	Happy	0.98
Happy	Neutral	0.6
Happy	Sad	0.05
Sad	Sad	0.98
Sad	Angry	0.46
Angry	Angry	0.98
Angry	Fear	0.14
Fear	Neutral	0.5
Boredom	Disgust	0.24
Disgust	Neutral	0.58
.....	.....	.....

Based on the psychological studies and experts' opinions, the above sets of emotions co-exist [2, 3]. Experts have provided the probabilities for their coexistence in speech segments.

3. Illustrating the co-existence of three emotions for speech segments

Emotion 1	Emotion 3	Emotion 2	Probability
Happy	Happy	Happy	0.98
Happy	Neutral	Happy	0.83
Happy	Neutral	Neutral	0.8
Neutral	Sad	Sad	0.73
Fear	Neutral	Fear	0.64
Sad	Boredom	Neutral	0.41
Fear	Fear	Angry	0.2
Sad	Fear	Boredom	0.3
Boredom	Happy	Neutral	0.1
Happy	Sad	Happy	0.1
.....	.....	.....	.....

Based on the psychological studies and experts' opinions, the above sets of emotions co-exist [2, 3]. Experts have provided the probabilities for their coexistence in speech segments.

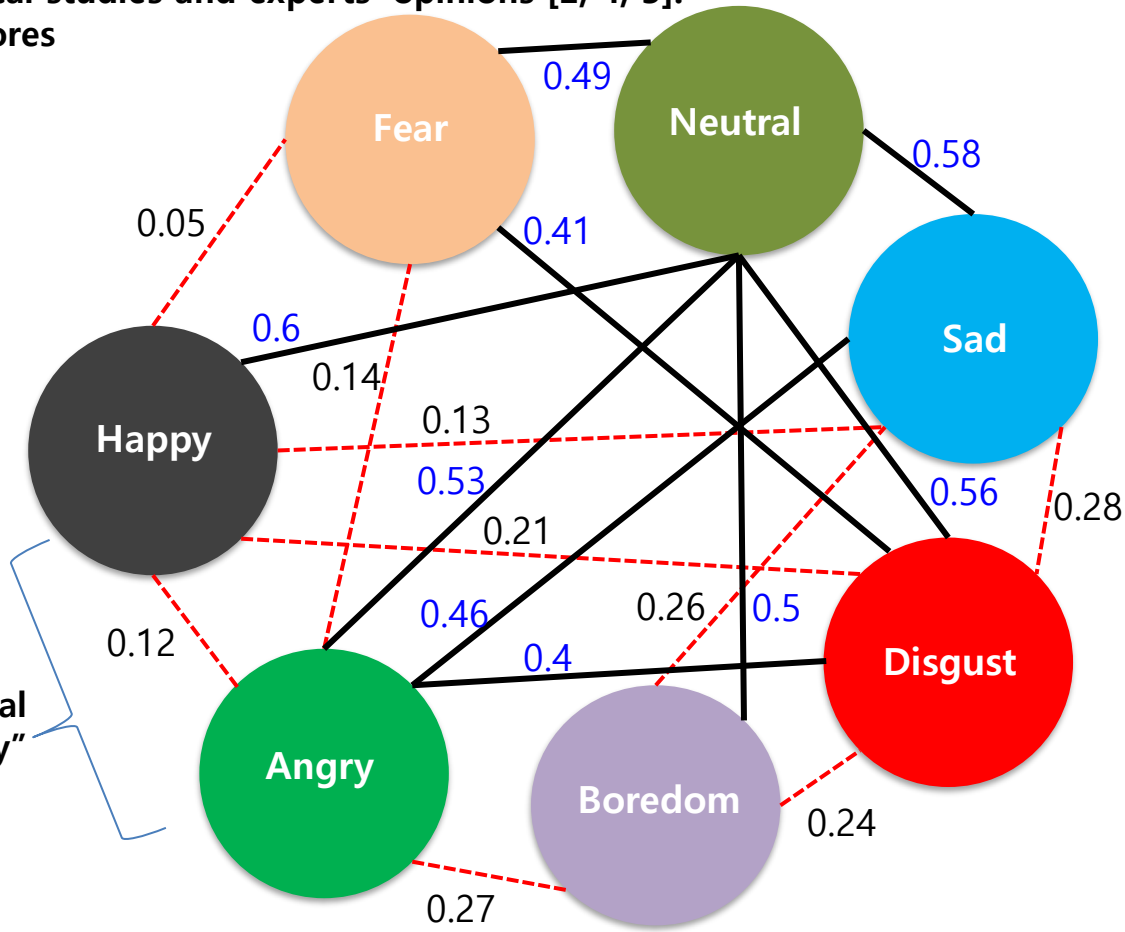
State transition diagram for conflicting and co-existent emotions

- Conflicting emotions refer to emotions that are mutually exclusive (i.e., both cannot occur together such as happy and sad).
- Co-existent emotions indicate emotions that are mutually inclusive (i.e., they can occur together such as neutral and happy).
- The diagram shows both conflicting and co-existent emotions with their probabilities.
- Probabilities are assigned based on psychological studies and experts' opinions [2, 4, 5].
- Conflicting emotions have lower probability scores
- Co-existent emotions have higher probabilities

Note:

Score — **Conflicting emotions**

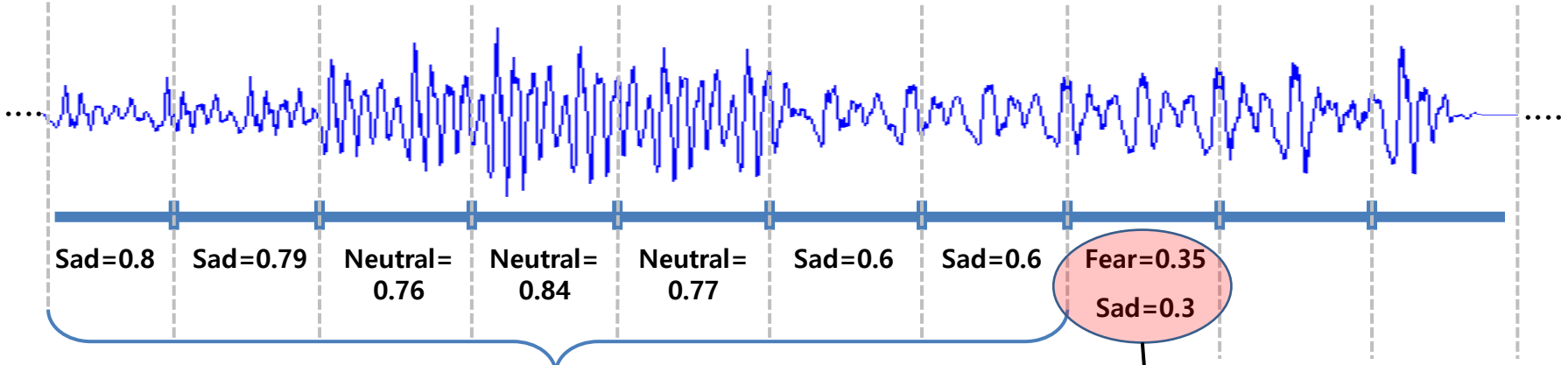
Score — **Co-existent emotions**



The probability that a speaker's emotional state can change from "Happy" to "Angry" state is 0.12.

4. Computing scores for previous N samples of a speech signal

Scenario 1



Suppose N=7, where N represents the number of previous speech segments.

Emotion 1: Sad=E1=4
Emotion 2: Neutral=E2=3

Confidence scores with small difference

$$\text{Emotion}_x \text{ score} = \frac{\text{No. of occurrences of Emotion}_x \text{ in N}}{\text{N}}$$

$$\text{Emotion 1: Sad} = \frac{4}{7} = 0.57$$

$$\text{Emotion 2: Neutral} = \frac{3}{7} = 0.43$$

Maximum = max (0.57 , 0.43)=0.57->Sad

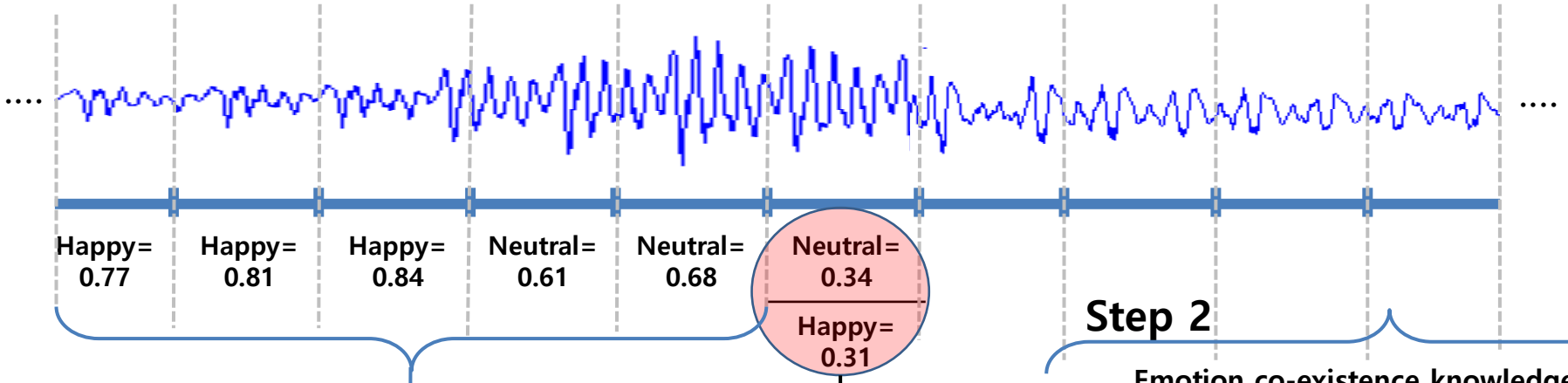
E 1
E 2
.
.
.
E x

= Number of detected emotions

The resultant score (0.57 in this scenario) will be combined with the original confidence score and confidence score computed from emotion co-existence. Based on the aggregated score, the emotion with maximum confidence score will be selected as a final output for a speech segment.

4. Computing scores for previous N samples and co-existence score for speech segments

Scenario 2



E1: Happy=3, E2: Neutral=2, N=5

Confidence scores with small difference

Step 1

$$\text{Emotion}_x \text{ score} = \frac{\text{No. of occurrences of Emotion}_x \text{ in N}}{\text{N}}$$

$$\text{E1: Happy} = \frac{3}{5} = 0.6$$

$$\text{E2: Neutral} = \frac{2}{5} = 0.4$$

Maximum = max (0.6 , 0.4)=0.6->Happy

Step 2

Emotion co-existence knowledge base

Emotion 1	Emotion 2	Emotion 3	Probability
Happy	Neutral	Neutral	0.39
Happy	Neutral	Happy	0.39
Sad	Sad	Neutral	0.37
.....	.....	.....	.....

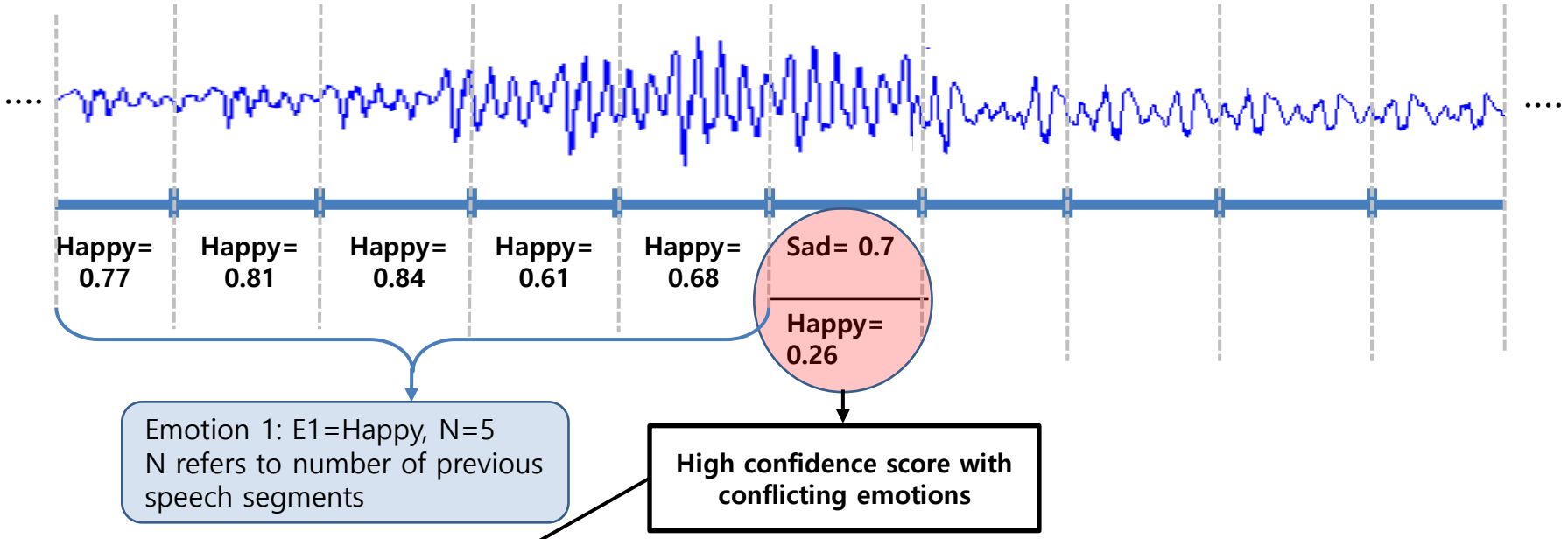
Case 1: Happy->Neutral->Neutral: 0.39

Case 2: Happy->Neutral->Happy: 0.39

The resultant score from step 1 is combined with individual cases of step 2 and original predicted confidence score, and the maximum resultant confidence value is returned. Based on the aggregated score, the emotion with maximum confidence score is selected as a final output for a speech segment.

5. Resolving conflict between conflicting emotions

Scenario 1



Steps:

1. Count the number of unique emotions in previous N samples (It is 1 in this case: Happy)
2. Check the probability score of different combinations of unique emotions of step 1 from emotion co-existence knowledge base [In this case: **(Happy, Sad)** and (**Happy, Happy**)]
3. Select the combination of unique emotions having maximum confidence score
4. Output the last emotional state of the selected combination of unique emotions

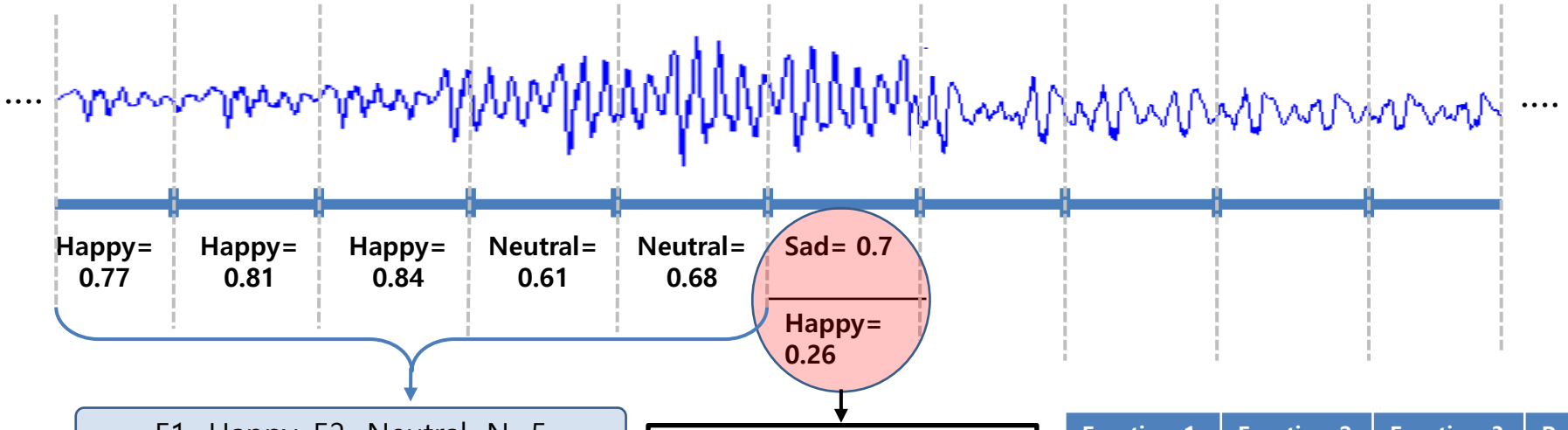
Case 1: Happy, Sad
Case 2: Happy, Happy

Emotion 1	Emotion 2	Probability
Happy	Happy	0.98
Happy	Neutral	0.7
Happy	Fear	0.34
Happy	Boredom	0.32
Happy	Sad	0.1
.....	.....	.....

As a result of above analysis, the predicted emotional state is Happy

5. Resolving conflict between conflicting emotions

Scenario 2



E1=Happy, E2=Neutral, N=5

High confidence score with conflicting emotions

Steps:

1. Count the number of unique emotions in previous N samples
   (**Unique emotions are 2 in this case, Happy and Neutral**)
2. Check the probability score of different combinations of unique emotions from emotion co-existence knowledge base
3. Select the combination of unique emotions having maximum confidence score
4. Output the last emotional state of the selected combination of unique emotions

Case 1: Happy, Neutral, Sad
Case 2: Happy, Neutral, Happy

Emotion 1	Emotion 2	Emotion 3	Probability
Happy	Happy	Happy	0.98
Happy	Neutral	Happy	0.5
Happy	Happy	Neutral	0.48
Happy	Neutral	Sad	0.2
Happy	Fear	Boredom	0.18
Happy	Sad	Sad	0.1
.....	.....	.....	.....

As a result of above analysis, the predicted emotional state is Happy

6. Reliability-Weighted fusion for confidence values of three different sources

Source 1 **Original Prediction**	Score 1 = 0.5-> Happy Score 2 = 0.58-> Neutral	**Source 2** **Speaker's History**	Score 1 = 0.6-> Happy Score 2 = 0-> Neutral	**Source 3** **Emotion Co-existence**	Score 1= 0.99->Happy Score 2= 0.79->Neutral

Problem: We need to combine the confidence values of three sources and decide the emotional state for a speech segment.
Solution: We propose reliability-weighted fusion to solve this problem. The formula of our fusion method is given as follows:

$$Fusion(Emotion\ State) = (r_1 S_1 + r_2 S_2 + r_3 S_3)$$

where r_1, r_2 and r_3 are reliability weights and S_1, S_2 and S_3 are confidence scores of different sources, respectively

Case 1 (Happy): (S_1=0.5, S_2=0.6, S_3=0.99)

$$Fusion(Emotion\ State) = (r_1 S_1 + r_2 S_2 + r_3 S_3)$$

Reliability weights

Original Prediction	Speaker's History	Emotion Co-existence
r_1	r_2	r_3
0.15	0.5	0.35

$$F(Happy) = (0.15 * 0.5 + 0.5 * 0.6 + 0.35 * 0.99)$$

$$F(Happy) = 0.7215$$

Case 2 (Neutral): (S_1=0.58, S_2=0, S_3=0.79)

$$Fusion(Emotion\ State) = (r_1 S_1 + r_2 S_2 + r_3 S_3)$$

Reliability weights

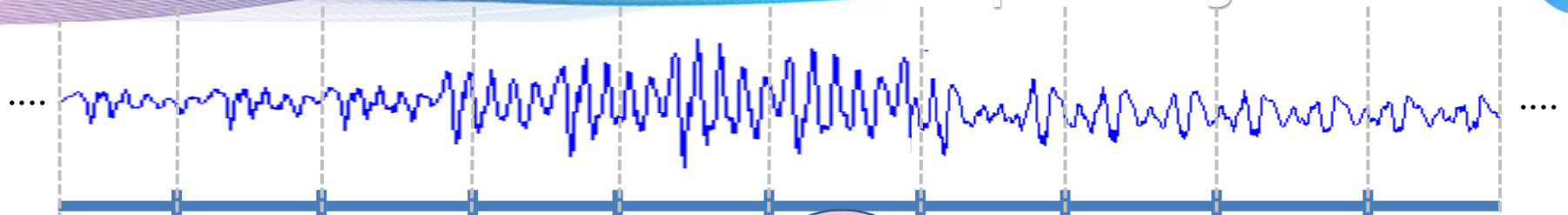
Original Prediction	Speaker's History	Emotion Co-existence
r_1	r_2	r_3
0.15	0.5	0.35

$$F(Neutral) = (0.15 * 0.58 + 0.5 * 0 + 0.35 * 0.79)$$

$$F(Neutral) = 0.3635$$

Final score = max (F(Happy), F(Neutral))= max (0.7215, 0.3635)= 0.7215->Fused score of "Happy" is greater than fused score of "Neutral", thus, "Happy" will be predicted as final emotion.

Scenario for computing scores of previous N samples and co-existence score and their fusion for speech segments



E1: Happy=4, E2: Neutral=1, N=5

- Conflicting emotions
- Original Predications

Step 1: compute a score for N samples

$$\text{Emotion}_x \text{ score} = \frac{\text{No. of occurrences of Emotion}_x \text{ in N}}{N}$$

$$\text{Happy} = \frac{4}{5} = 0.8$$

$$\text{Neutral} = \frac{1}{5} = 0.2$$

Maximum = max (0.8 , 0.2)=**0.8**->Happy

Step 2: Resolve conflict

Emotion co-existence knowledge base

Emotion 1	Emotion 2	Emotion 3	Probability
Happy	Neutral	Neutral	0.72
Happy	Neutral	Happy	0.7
Happy	Neutral	Sad	0.2
.....	.....	.....	.....

Case 1: Happy->Neutral->Sad: 0.2

Case 2: Happy->Neutral->Happy: 0.7

Step 3: Fusion

Combine the score of step 1 (o.8 in this scenario) to each confidence score of different cases of step 2 and original prediction and decide about sad or happy

Case 1

$$F(Sad) = (0.15*0.34 + 0.5*0 + 0.35*0.2)$$

$F(Sad) = 0.121$ **0.121** is the fused confidence score for sad state

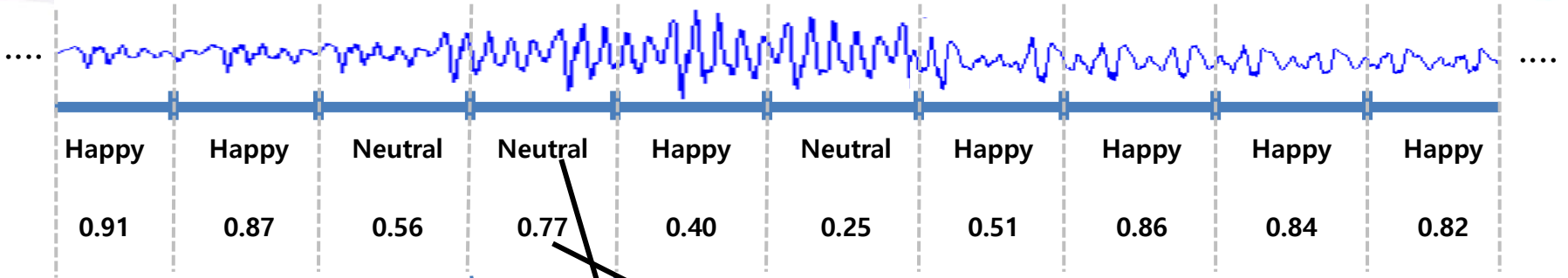
Case 2

$$F(Happy) = (0.15*0.31 + 0.5*0.8 + 0.35*0.7)$$

$F(Happy) = 0.691$ **0.691** is the confidence score for happy

Final score = max (F(Sad)), F(Happy))= max (0.121, 0.691)= 0.691->Score of sad is less than happy, thus, happy will be outputted as final label

7. Maintaining Emotional History for a Speaker



Speech Segment	Emotion	Probability
1	Happy	0.91
2	Happy	0.87
3	Neutral	0.56
4	Neutral	0.77
5	Happy	0.40
6	Neutral	0.25
7	Happy	0.51
8	Happy	0.86
.....	.....	.....

Overall emotional state: Happy

7. Maintaining Emotional History for a Speaker

Speech Segment	Emotion	Probability
1	Fear	0.90
2	Fear	0.83
3	Neutral	0.57
4	Fear	0.87
5	Neutral	0.44
6	Fear	0.52
7	Fear	0.57
8	Fear	0.80
.....	.....	.....

Overall emotional state: Fear

Speech Segment	Emotion	Probability
1	Angry	0.83
2	Angry	0.77
3	Fear	0.62
4	Disgust	0.58
5	Angry	0.64
6	Disgust	0.48
7	Angry	0.51
8	Angry	0.68
.....	.....	.....

Overall emotional state: Angry

Our system maintains the emotional history of the speaker for better reasoning and validation of emotions for upcoming speech segments. The emotion history can also be used to determine the overall emotional state of a speaker for the duration of the call for further necessary operations at call centres.

7. Maintaining Emotional History for a Speaker

Speech Segment	Emotion	Probability
1	Sad	0.95
2	Sad	0.92
3	Sad	0.86
4	Boredom	0.75
5	Boredom	0.60
6	Sad	0.40
7	Sad	0.51
8	Sad	0.86
.....	.....	.....

Overall emotional state: Sad

Speech Segment	Emotion	Probability
1	Boredom	0.73
2	Sad	0.53
3	Boredom	0.61
4	Boredom	0.65
5	Boredom	0.68
6	Disgust	0.52
7	Sad	0.39
8	Boredom	0.73
.....	.....	.....

Overall emotional state: Boredom

Our system maintains the emotional history of the speaker for better reasoning and validation of emotions for upcoming speech segments. The emotion history can also be used to determine the overall emotional state of a speaker for the duration of the call for further necessary operations at call centres.

References

1. Ji, Shuiwang, Wei Xu, Ming Yang, and Kai Yu. "3D convolutional neural networks for human action recognition." IEEE transactions on pattern analysis and machine intelligence 35, no. 1 (2013): 221-231.
2. Wang, Shangfei, Jun Wang, Ziheng Wang, and Qiang Ji. "Multiple emotion tagging for multimedia data by exploiting high-order dependencies among emotions." IEEE Transactions on Multimedia 17, no. 12 (2015): 2185-2197.
3. https://www.psychologytoday.com/blog/evolution-the-self/201406/can-your-body-express-multiple-emotions-simultaneously (visited: 9th March 2026)
4. https://www.psychologytoday.com/blog/evolution-the-self/201406/can-you-feel-two-emotions-once (visited: 9th March 2026)
5. Williams, Patti, and Jennifer L. Aaker. "Can mixed emotions peacefully coexist?." Journal of Consumer Research 28, no. 4 (2002): 636-649.
6. Badshah, Abdul Malik, Nasir Rahim, Noor Ullah, Jamil Ahmad, **Khan Muhammad**, Mi Young Lee, Soonil Kwon, and Sung Wook Baik. "Deep features-based speech emotion recognition for smart affective services." Multimedia Tools and Applications 78, no. 5 (2019): 5571-5589.
7. APPARATUS AND METHOD FOR SPEECH EMOTION RECONGNITION USING A REASONING PROCESS (추론 프로세스를 이용한 음성 감정 인식 장치 및 방법), Korea Intellectual Property Rights Information Service (KIPRIS).







Thank you
Q&A







